

VASANTHAKUMAR S

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PROFILE SUMMARY

Innovative and driven Electrical and Electronics Engineering undergraduate (GPA 8.1/10) with a strong cross-disciplinary foundation spanning embedded hardware, robotics, and modern full-stack web development. Proven ability to take projects from concept to working prototype from ultrasonic sensor arrays integrated with ROS to AI-powered voice interview platforms built on locally hosted LLMs. Comfortable working across the full stack (React, Node.js, MySQL) as well as at the hardware level (Arduino, Embedded C). Passionate about building practical, real-world solutions and eager to contribute to a team tackling challenging engineering or software problems.

EDUCATION

B.E. – Electrical and Electronics Engineering

2023 – 2027 (Expected)

Sri Manakula Vinayagar Engineering College, Puducherry · GPA: 8.1 / 10.0

- Consistently maintained a GPA above 8.0 across all semesters, ranking among the top performers in the department.
- Actively involved in technical clubs and project-based learning, applying classroom concepts to hands-on hardware and software builds.
- Pursuing coursework spanning Power Systems, Microprocessors & Microcontrollers, Control Systems, Digital Electronics, and Signal Processing.

HSC – Computer Science

2023

Aditya Vidyashram Residential School · Aggregate: 85.4%

- Completed higher secondary education with Computer Science as a core subject, building early programming foundations in C++ and Python.
- Developed strong problem-solving and analytical skills through mathematics and science coursework, forming the basis for engineering study.

INTERNSHIP EXPERIENCE

Web Development Intern — Cloud Logic

Dec 2025

Puducherry, India · Short-Term Industry Exposure Programme

- Gained hands-on exposure to the full web development lifecycle, from requirement understanding through UI component development and testing.
- Built and tested responsive front-end UI components using HTML, CSS, and JavaScript, learning industry-standard coding practices and workflows.
- Collaborated with senior developers to understand real-world project structure, version control practices, and deployment considerations.
- Strengthened understanding of client-server architecture and the role of front-end technologies in larger web application ecosystems.

PROJECTS

Sonar Navigation Mapping System

2025

Stack: Arduino Uno · HC-SR04 Ultrasonic Sensors · Embedded C · ROS (Robot Operating System) · Serial Communication

- Conceptualized and built a fully functional low-cost 2D environmental mapping system using ultrasonic sensors mounted on a servo-driven Arduino platform, achieving real-time obstacle detection and spatial awareness.
- Developed custom Embedded C firmware for the Arduino to control servo sweep angles, trigger ultrasonic pulses, and transmit distance readings via serial at 115200 baud to a host machine.
- Integrated the hardware data stream with ROS nodes in Python to parse, transform, and publish sensor data as a ROS topic, enabling real-time occupancy grid map visualization in RViz.
- Achieved sub-200ms map refresh latency across a 180° sweep range, demonstrating system viability as a low-cost alternative to LIDAR-based navigation in constrained environments.
- Documented system architecture, wiring schematics, and ROS node graph for future extensibility and academic presentation.

Interview Assistant AI Agent

2026

- **Stack:** Ollama (LLaMA 3) · Node.js · Express.js · React.js · MySQL · Coqui/Piper TTS · HTML · CSS · JavaScript · REST API
- Designed and developed a full-stack AI-powered interview simulation platform that conducts real-time, voice-driven technical interviews entirely on local infrastructure — ensuring zero cloud dependency and complete data privacy.
- Integrated Ollama (LLaMA 3) as the AI reasoning engine to dynamically generate role-specific technical questions, evaluate candidate responses, and produce personalized feedback and career guidance in natural language.
- Implemented Coqui/Piper TTS for natural, human-like audio question delivery and built a React.js front-end with real-time speech recognition for seamless voice-based candidate interaction.
- Architected a Node.js/Express.js REST API backend with MySQL integration for persistent session management, interview history logging, and candidate performance analytics.
- Applied prompt engineering techniques to fine-tune LLaMA 3 behaviour for structured interview formats across domains including DSA, system design, and core engineering subjects.

TECHNICAL SKILLS

Programming	Python · Java · Embedded C
Web & Full-Stack	React.js · Node.js · Express.js · HTML5 · CSS3 · REST APIs
Embedded & Hardware	Arduino Uno/Nano · Ultrasonic / Servo / IR Sensors · Serial Communication
Simulation & Design	Fusion 360 · MATLAB · Proteus · Autodesk (AutoCAD / Inventor)
Databases & AI/ML	MySQL · Ollama · LLaMA 3 · Prompt Engineering · TTS (Coqui/Piper)
Robotics & OS	ROS (Robot Operating System) · RViz · Windows · Linux (Ubuntu)
Dev Tools	Git · VS Code · Arduino IDE · Postman · npm
Soft Skills	Team Collaboration · Public Speaking · Creative Problem-Solving · Self-Learning

KEY COMPETENCIES

Embedded System Design · Hardware-Software Integration · Real-Time Systems · Full-Stack Web Development · AI Application Development · Sensor Fusion · ROS-Based Robotics · Prompt Engineering · Rapid Prototyping · Technical Documentation · Agile Mindset · Cross-Functional Collaboration

CERTIFICATIONS

- Java Basics – Programming Course (Hands-On) · SkillRack · 2024
- Python Basics – Programming Course (Hands-On) · SkillRack · 2024
- Web Development Fundamentals – Industry Training · Cloud Logic · Dec 2025

RELEVANT COURSEWORK

Microprocessors & Microcontrollers · Digital Electronics & Logic Design · Control Systems · Power Electronics · Signal Processing · Electric Circuits & Networks · Data Structures & Algorithms · Object-Oriented Programming · Computer Networks (elective)

EXTRACURRICULARS & ACHIEVEMENTS

- Active participant in department-level technical fests and project expo competitions, presenting embedded systems projects to faculty panels and industry judges.
- Demonstrated strong public speaking and presentation skills through multiple intra-college technical seminars and group project demos.
- Self-taught full-stack web development and AI integration beyond curriculum, independently building production-grade projects using React, Node.js, and local LLMs.
- Consistently recognized for creative problem-solving and collaborative attitude in team-based project environments.

DECLARATION

I hereby declare that all information provided in this resume is true and accurate to the best of my knowledge. I am available for interviews at your convenience and look forward to contributing to your organization.

Place: Puducherry

Vasanthakumar